Spare parts stock policy audit

Sample finding — one page, produced from two years of part movement history

What this example is based on

Open benchmark data: 2,674 automotive part numbers, 51 months of monthly movement (Monash Time Series Forecasting Archive, CC-BY 4.0). No client data was used.

75.9% **of all month-part observations are zero.**
Average-consumption logic is answering a different question.

99.4% **of parts fall outside the 'smooth' region.**
One model applied to the whole catalogue is an unearned claim.

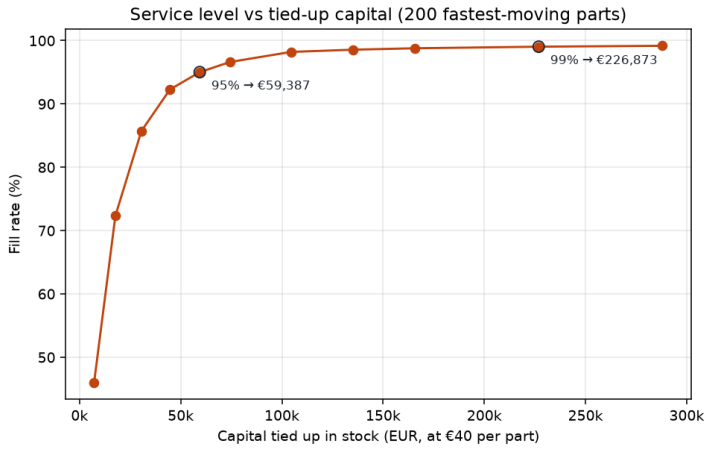
16.6% **lower forecast error than the ERP baseline.**
Achieved by selecting the method per part, validated out-of-sample.

698 **parts had no movement in the last 12 months.**
A cash-release list that needs no forecasting at all.

The decision this produces

Going from 95% to 99% fill rate costs five times the capital.

95% fill = EUR 59,400 tied up · 99% fill = EUR 288,000 tied up (200 fastest-moving parts, 2-month lead time)



To draw this curve with your own part numbers

One CSV export from your ERP: part number, date, quantity moved. Two to three years.
Optional: unit cost and lead time. No sensors, no integration, no pilot server.
Turnaround for one part family: 2 working days, free of charge.

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Method and code: github.com/yigitckk/intermittent-demand-audit